

GORNA ORYAHOVITSA, Bulgaria

WMO: 155253

Lat: 43.151N

Lon: 25.713E

Elev: 285

StdP: 14.55

Time Zone: 2.00 (EUE)

Period: 97-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	5.3	12.0	3.0	6.5	6.6	8.7	8.7	12.7	26.6	39.9	23.6	40.3	1.7	90	0.483

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	25.7	95.1	70.5	91.7	70.3	89.3	69.6	74.7	86.8	73.0	84.9	71.6	83.3	6.7	300

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
71.5	117.4	80.1	69.7	110.4	77.8	68.0	104.1	76.1	38.4	87.0	36.9	85.3	35.6	84.2	81.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
22.0	18.3	15.1	-3.0	99.7	7.1	4.1	-8.1	102.7	-12.2	105.0	-16.2	107.3	-21.4	110.3		
			-2.9	77.1	7.3	2.0	-8.1	78.5	-12.4	79.7	-16.4	80.8	-21.7	82.3		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	53.6	30.8	36.3	44.6	53.7	62.5	69.8	73.6	73.1	64.3	54.2	45.1	34.5
	DBStd	16.44	10.28	10.49	7.84	6.51	5.33	5.20	4.60	4.70	6.06	7.49	8.11	9.44
	HDD50	1949	598	390	204	34	1	0	0	0	0	45	191	486
	HDD65	4942	1061	805	632	341	110	16	2	3	86	341	599	946
	CDD50	3269	2	5	36	146	388	593	731	717	428	174	44	5
	CDD65	786	0	0	0	3	32	159	268	254	64	5	1	0
	CDH74	10149	0	0	10	120	644	1916	3204	3136	964	148	7	0
	CDH80	4374	0	0	0	20	145	770	1536	1532	341	30	0	0
Wind	WSAvg	5.3	4.8	5.7	6.8	6.4	5.5	5.3	5.6	4.8	5.0	4.5	4.8	4.7
Precipitation	PrecAvg	25.6	1.7	1.7	1.9	2.1	3.1	2.7	2.6	1.9	2.5	1.8	1.6	2.2
	PrecMax	44.3	4.0	3.5	6.0	4.4	6.6	5.4	8.0	5.5	8.7	4.5	3.8	3.8
	PrecMin	16.7	0.4	0.2	0.1	0.1	0.9	0.4	0.2	0.2	0.1	0.0	0.1	0.3
	PrecStd	6.4	0.8	0.9	1.3	1.1	1.7	1.6	2.1	1.4	2.0	1.1	1.0	0.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	62.3	68.3	75.2	82.5	87.7	95.2	102.0	98.8	93.1	84.1	74.9	64.6
		MCWB	52.7	54.8	56.7	63.6	67.1	71.4	70.7	69.7	67.2	64.6	60.9	54.0
	2%	DB	55.8	62.6	70.0	77.3	84.0	91.3	95.4	95.0	87.9	78.4	69.7	58.9
		MCWB	48.7	52.0	54.3	60.9	65.5	71.8	70.0	70.7	66.0	62.5	59.2	50.8
	5%	DB	50.2	57.4	66.0	72.0	80.7	87.8	91.7	91.6	84.0	73.4	64.7	53.7
		MCWB	45.5	49.4	52.5	57.9	64.6	70.4	70.2	70.2	65.2	61.2	56.4	48.0
	10%	DB	44.6	52.0	61.0	68.3	77.3	84.4	88.1	88.2	80.2	68.4	59.4	48.1
		MCWB	41.8	45.9	50.3	55.9	63.7	69.2	69.6	68.7	64.4	59.0	53.4	44.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	53.2	55.7	58.2	65.6	70.6	76.4	76.3	77.9	71.0	67.5	61.7	55.3
		MCDB	61.2	65.3	69.7	79.4	81.9	88.6	89.0	89.5	83.0	79.5	71.7	63.5
	2%	WB	49.6	52.4	55.4	62.4	68.3	73.9	74.4	74.6	68.8	64.3	59.3	51.3
		MCDB	55.9	61.2	67.4	75.0	78.7	85.8	87.0	86.2	80.9	74.0	68.3	57.4
	5%	WB	45.8	49.8	53.2	59.6	66.6	72.2	72.8	72.6	67.0	62.0	57.0	48.6
		MCDB	49.5	57.7	64.2	69.5	77.5	83.7	85.0	84.3	79.4	71.3	63.8	53.5
	10%	WB	41.6	46.5	50.8	57.4	64.8	70.5	71.5	70.8	65.5	59.7	54.3	44.9
		MCDB	44.1	51.3	60.1	66.7	74.8	81.3	83.0	83.1	76.9	67.3	59.5	47.2
Mean Daily Temperature Range	5% DB	MDBR	16.3	19.5	22.2	24.0	24.7	24.5	25.7	27.4	25.2	23.3	18.2	16.3
		MCDBR	25.8	30.1	33.3	31.5	31.6	30.2	33.2	33.3	33.8	32.7	28.1	26.7
		MCWBR	18.6	19.5	18.4	17.1	15.6	12.9	12.0	12.0	14.0	18.2	17.8	19.1
	5% WB	MCDBR	24.4	28.8	29.5	27.7	27.1	27.1	26.5	27.3	27.8	28.8	25.4	24.3
		MCWBR	18.0	19.1	17.0	15.8	14.9	13.2	11.9	12.0	13.3	16.7	16.5	17.7
Clear-Sky Solar Irradiance	taub	0.350	0.384	0.429	0.487	0.461	0.467	0.474	0.487	0.453	0.410	0.376	0.352	
	taud	2.273	2.192	2.114	1.967	2.114	2.141	2.141	2.131	2.088	2.147	2.246	2.293	2.279
	Ebn at Noon	240	251	254	248	259	258	254	246	244	239	229	226	
	Edh at Noon	28	36	44	55	49	48	48	48	42	34	28	26	
All-Sky Solar Radiation	RadAvg	482	733	1079	1485	1840	2014	2058	1851	1355	881	526	401	
	RadStd	69	94	97	158	147	147	147	109	106	94	67	54	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (41 neighbors)	+0.79	N/A	N/A	N/A	+0.77	N/A	N/A	-230	+151	N/A

Nomenclature: See separate page